



PENG SAMNANG

Full-Stack Developer

EDUCATION

2025 - Present

- Bachelor of Computer Science At Cambodia University of Management and Technology

2021.27. December

- Graduated at Thmor Kre High School

TECHNICAL SKILLS

- **Frontend:** HTML5, CSS3, JavaScript, TypeScript, React.js, Tailwind CSS, Vite
- **Backend:** Python, Django, Django REST Framework, REST APIs, JWT Authentication
- **Database:** PostgreSQL, MySQL
- **Tools:** Git, GitHub, Figma

LANGUAGES

- English
- Khmer

CONTACT

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PROFILE

Full-Stack Web Developer with hands-on experience building web applications using React, Django, and PostgreSQL. Skilled in REST API development, responsive frontend implementation, database design, and full-stack system integration. Passionate about building scalable applications that solve real-world business problems.

EXPERIENCE

WEB DEVELOPER (FREELANCE) 2025 - Present

Remote / Local Clients

- Build and customize web applications based on client business requirements.
- Develop frontend interfaces and backend APIs for modern web systems.
- Work on responsive UI, database integration, and deployment-ready solutions.

Junior Python Backend Developer AT KILO IT 2024 - 2025

- Developed REST APIs using Django REST Framework.
- Assisted in database design and backend business logic implementation.
- Worked with Python and Django to build backend services.

PROJECTS

LOGISTIC DELIVERY SYSTEM

- BUILT A FULL-STACK LOGISTICS MANAGEMENT SYSTEM USING REACT, DJANGO REST FRAMEWORK, AND POSTGRESQL.
- IMPLEMENTED SHIPPING ORDERS, INVOICE TRACKING, PAYMENTS, REPORTS, QR WORKFLOWS, AND PUBLIC ORDER LOOKUP.

SWIMMING COURSE WEBSITE

- BUILT A RESPONSIVE WEBSITE FOR PRESENTING COURSE INFORMATION AND USER-FACING CONTENT.

UNIVERSITY MANAGEMENT SYSTEM (ONGOING)

- DEVELOPING A UNIVERSITY MANAGEMENT SYSTEM FOR ACADEMIC AND ADMINISTRATIVE WORKFLOWS.

AI FACE RECOGNITION

PROJECT NAME: LOGISTIC DELIVERY SYSTEM

PROJECT TYPE: FULL-STACK LOGISTICS MANAGEMENT SYSTEM

PROJECT OVERVIEW:

LOGISTICSYS IS A FULL-STACK WEB APPLICATION DEVELOPED FOR MANAGING SHIPPING ORDERS, INVOICES, PAYMENTS, REPORTS, AND QR-BASED LOGISTICS WORKFLOWS. THE SYSTEM SUPPORTS BOTH ADMINISTRATORS AND CUSTOMERS THROUGH ROLE-BASED ACCESS CONTROL AND PROVIDES PUBLIC ORDER LOOKUP FUNCTIONALITY.

RESPONSIBILITIES:

- DEVELOPED BACKEND APIS USING DJANGO REST FRAMEWORK
- BUILT FRONTEND INTERFACES USING REACT, TYPESCRIPT, AND TAILWIND CSS
- MPLEMENTED JWT AUTHENTICATION AND SESSION HANDLING
- DEVELOPED ORDER, INVOICE, PAYMENT, AND REPORTING MODULES
- BUILT QR CODE SCANNING AND IMAGE-BASED QR PROCESSING WORKFLOWS
- MPLEMENTED PUBLIC TRACKING/SEARCH BY INVOICE NUMBER
- DEVELOPED USER MANAGEMENT AND CUSTOM PERMISSION CONTROL

KEY FEATURES:

- SHIPPING ORDER CREATION AND STATUS TRACKING
- AUTOMATIC INVOICE GENERATION
- PAYMENT MANAGEMENT AND REVENUE DASHBOARD
- REPORT FILTERING AND CSV/PDF EXPORT
- QR ORDER WORKFLOW AND IMAGE UPLOAD
- PUBLIC ORDER/INVOICE LOOKUP
- ADMIN AND CUSTOMER ROLE MANAGEMENT

TECHNOLOGIES USED:

- FRONTEND: REACT, TYPESCRIPT, VITE, TAILWIND CSS
- BACKEND: DJANGO, DJANGO REST FRAMEWORK, SIMPLE JWT
- DATABASE: POSTGRESQL
- DEPLOYMENT: RENDER, VERCEL

PROJECT NAME: RITHYA SWIMMING SCHOOL WEBSITE

PROJECT TYPE : FULL-STACK CONTENT MANAGEMENT WEBSITE

PROJECT OVERVIEW:

RITHYA IS A FULL-STACK WEB APPLICATION DEVELOPED FOR A SWIMMING SCHOOL TO MANAGE AND DISPLAY PUBLIC WEBSITE CONTENT. THE SYSTEM INCLUDES A PUBLIC-FACING WEBSITE FOR VISITORS AND AN ADMIN DASHBOARD FOR MANAGING HERO BANNERS, ABOUT CONTENT, PROGRAMS, GALLERY IMAGES, TESTIMONIALS, AND CONTACT INFORMATION.

RESPONSIBILITIES:

- DEVELOPED BACKEND APIS USING DJANGO REST FRAMEWORK
- BUILT FRONTEND INTERFACES USING REACT AND VITE
- IMPLEMENTED TOKEN-BASED AUTHENTICATION FOR ADMIN ACCESS
- DEVELOPED ADMIN DASHBOARD FOR CONTENT MANAGEMENT
- INTEGRATED CLOUDINARY FOR IMAGE UPLOAD AND MEDIA MANAGEMENT
- BUILT PUBLIC API ENDPOINTS FOR WEBSITE SECTIONS
- PREPARED DEPLOYMENT CONFIGURATION FOR PRODUCTION ENVIRONMENT

KEY FEATURES:

- PUBLIC WEBSITE FOR ABOUT, PROGRAMS, GALLERY, TESTIMONIALS, AND CONTACT INFORMATION
- ADMIN DASHBOARD WITH CRUD OPERATIONS FOR WEBSITE CONTENT
- HERO BANNER MANAGEMENT
- PROGRAM AND GALLERY MANAGEMENT
- TESTIMONIAL AND CONTACT INFORMATION MANAGEMENT
- TOKEN-BASED ADMIN LOGIN/LOGOUT
- IMAGE UPLOAD AND STORAGE WITH CLOUDINARY

TECHNOLOGIES USED:

- FRONTEND: REACT, VITE, TAILWIND CSS
- BACKEND: DJANGO, DJANGO REST FRAMEWORK
- DATABASE: POSTGRESQL
- DEPLOYMENT: RENDER
- MEDIA STORAGE: CLOUDINARY

PROJECT NAME: UNIVERSITY MANAGEMENT SYSTEM (UMS)

PROJECT TYPE: FULL-STACK UNIVERSITY MANAGEMENT SYSTEM

PROJECT OVERVIEW:

UMS IS A FULL-STACK WEB APPLICATION DEVELOPED TO MANAGE ACADEMIC AND ADMINISTRATIVE OPERATIONS IN A UNIVERSITY ENVIRONMENT. THE SYSTEM SUPPORTS MULTIPLE USER ROLES INCLUDING ADMINISTRATORS, STAFF, LECTURERS, AND STUDENTS, AND PROVIDES MODULES FOR ACADEMIC STRUCTURE MANAGEMENT, SEMESTER LIFECYCLE CONTROL, TIMETABLE SCHEDULING, ENROLLMENT, ATTENDANCE, GRADES, LECTURER APPLICATIONS, TEACHING CONTRACTS, AND CUSTOM ROLE-PERMISSION MANAGEMENT.

RESPONSIBILITIES:

- DEVELOPED BACKEND APIS USING DJANGO REST FRAMEWORK
- BUILT FRONTEND INTERFACES USING REACT, TYPESCRIPT, VITE, AND TAILWIND CSS
- IMPLEMENTED TOKEN-BASED AUTHENTICATION AND ROLE-BASED ROUTING
- DEVELOPED ACADEMIC STRUCTURE MODULES FOR FACULTIES, DEPARTMENTS, MAJORS, SUBJECTS, AND CLASS GROUPS
- IMPLEMENTED ACADEMIC YEAR AND SEMESTER STATE MANAGEMENT WORKFLOWS
- BUILT TIMETABLE AND SCHEDULE MANAGEMENT FEATURES
- DEVELOPED STUDENT ENROLLMENT, ATTENDANCE, AND GRADE MANAGEMENT MODULES
- BUILT LECTURER APPLICATION, TEACHER PROFILE, AND TEACHING CONTRACT WORKFLOWS
- IMPLEMENTED CUSTOM ROLE AND PERMISSION MANAGEMENT FOR SYSTEM ADMINISTRATION

KEY FEATURES:

- MULTI-ROLE ACCESS FOR ADMIN, STAFF, LECTURER, AND STUDENT USERS
- FACULTY, DEPARTMENT, MAJOR, SUBJECT, AND CLASS GROUP MANAGEMENT
- ACADEMIC YEAR AND SEMESTER STATUS CONTROL WITH CURRENT/ACTIVE PERIOD HANDLING
- TIMETABLE AND SCHEDULE BUILDER FOR CLASSES, TEACHERS, ROOMS, AND SESSIONS
- STUDENT ENROLLMENT, ATTENDANCE TRACKING, AND GRADE MANAGEMENT
- LECTURER APPLICATION APPROVAL AND TEACHING CONTRACT MANAGEMENT
- TEACHER ATTENDANCE AND STAFF ACTIVITY TRACKING
- CUSTOM ROLE AND PERMISSION ASSIGNMENT
- SYSTEM SETTINGS, REPORTING, AND ACADEMIC OPERATIONS DASHBOARD

TECHNOLOGIES USED:

- FRONTEND: REACT, TYPESCRIPT, VITE, TAILWIND CSS, SHADCN/UI
- BACKEND: DJANGO, DJANGO REST FRAMEWORK, TOKEN AUTHENTICATION
- DATABASE: POSTGRESQL
- DEPLOYMENT: RENDER, VERCEL

PROJECT NAME: AI FACE RECOGNITION ATTENDANCE SYSTEM

PROJECT TYPE: AI-POWERED ATTENDANCE MANAGEMENT SYSTEM

PLATFORM: WEB APPLICATION

PROJECT OVERVIEW

FACTORY ATTENDANCE SYSTEM IS AN AI-POWERED EMPLOYEE ATTENDANCE MANAGEMENT PLATFORM DESIGNED FOR FACTORY ENVIRONMENTS. IT USES FACE RECOGNITION TO IDENTIFY EMPLOYEES THROUGH LIVE CAMERAS AND AUTOMATICALLY RECORD CHECK-IN AND CHECK-OUT EVENTS.

THE SYSTEM SUPPORTS:

- EMPLOYEE AND DEPARTMENT MANAGEMENT
- SHIFT AND BREAK-TIME MANAGEMENT
- REAL-TIME ATTENDANCE TRACKING
- OVERTIME AND LEAVE MANAGEMENT
- LIVE CAMERA MONITORING
- MOBILE SELF-SERVICE ATTENDANCE LOOKUP
- EXCEL AND PDF REPORTING

MY ROLE AND CONTRIBUTIONS

I CONTRIBUTED TO THE DEVELOPMENT OF THE FOLLOWING CORE FEATURES:

- DEVELOPED BACKEND APIS USING DJANGO REST FRAMEWORK
- IMPLEMENTED REAL-TIME CAMERA STREAMING AND WEBSOCKET COMMUNICATION
- BUILT THE FACE RECOGNITION WORKFLOW USING INSIGHTFACE, FAISS, AND OPENCV
- DEVELOPED SHIFT-BASED ATTENDANCE LOGIC FOR AUTOMATIC CHECK-IN/CHECK-OUT AND LATE DETECTION
- BUILT OVERTIME CALCULATION AND LEAVE MANAGEMENT FEATURES
- DEVELOPED DASHBOARD, EMPLOYEE MANAGEMENT, CAMERA MONITORING, AND MOBILE SELF-SERVICE PAGES
- IMPLEMENTED EXCEL/PDF REPORT EXPORT
- INTEGRATED TELEGRAM ALERTS FOR UNKNOWN FACE AND LATE ATTENDANCE EVENTS

Core Features

Employee and Department Management

- Create and manage employees and departments
- Store employee ID, department, shift, position, and hourly rate
- Register employee faces for recognition

Multi-Angle Face Registration

- Capture multiple face angles per employee
- Improve face matching accuracy through multi-angle embeddings
- Register faces directly through webcam capture on the frontend

Live Attendance Detection

- Support webcam and RTSP/IP cameras
- Identify employees from live video streams
- Deliver real-time recognition results through WebSockets
- Show live attendance feed on the dashboard

Shift and Attendance Logic

- Define shift times, grace periods, break time, and scan windows
- Detect `check\_in`, `check\_out`, `break`, and `ignore` events
- Calculate late minutes and duplicate scan cooldowns
- Automatically calculate work hours, undertime, and overtime

Overtime and Leave Management

- Create and manage overtime records
- Approve or reject overtime requests
- Calculate overtime pay from hourly rate
- Manage leave requests

Reporting

- Daily attendance summary
- Shift summary
- Overtime summary
- Excel and PDF export

Mobile Self-Service

- Employees can enter their employee ID to view their own attendance
- View today's attendance status, records, and monthly overtime

Alerts and Notifications

- Telegram alerts for unknown face detection
- Notifications for late attendance events

Technologies Used

Frontend

- - React
- - Axios
- - React Router
- - react-webcam
- - react-hot-toast
- - Recharts

Backend

- Python
- Django
- Django REST Framework
- Django Channels
- django-filter
- Celery Beat

AI / Computer Vision

- InsightFace
- FAISS
- OpenCV
- NumPy

Database / Queue / Deployment

- PostgreSQL / MySQL
- Redis
- Docker / Docker Compose

Technical Highlights

- Built an AI-powered attendance system using face recognition and real-time camera streaming
- Used FAISS for scalable face similarity search across large employee datasets
- Implemented shift-based business logic for automatic check-in/check-out, late detection, and overtime calculation
- Developed WebSocket consumers for live attendance updates
- Built Excel/PDF reporting and mobile self-service features

Skills Demonstrated

- Full-stack web development
- REST API development
- Real-time systems with WebSocket
- Face recognition integration
- Computer vision workflows
- Attendance and shift business logic
- Reporting and export features